GlobalGeoTree — Climate–Family Dependencies

Direct exploration: which families prefer which climates, and where (GGT-10kEval-90)

0. Why this document is separate from the PCA/t-SNE one

The earlier exploration leaned on PCA and t-SNE to find structure automatically. On this dataset that approach hit a ceiling: `level0` (deciduous/evergreen broadleaf/evergreen needleleaf) doesn't separate cleanly in unsupervised projections, partly because it's a coarse 3-class label and partly because compressed components (PC1, PC2, t-SNE/UMAP axes) are hard to map back to "this family likes cold places."

This document asks the question directly instead: **for specific families, do climate and elevation values differ in the way ecological intuition suggests** (e.g. conifers skewing colder/higher, certain families skewing warmer/drier)? Every section below uses real, interpretable variables (°C, mm/yr, meters) rather than abstract components, and ends in either a table or a map you can sanity-check against known geography.

▼ Code

```
library(jsonlite)
library(dplyr)
library(purrr)
library(tibble)
library(stringr)
library(ggplot2)
library(tidyr)
library(forcats)

theme_set(theme_minimal())
```

1. Loading the data

Same robust-by-name loader as before — set `root` to your local extracted `GGT-10kEval-90` folder and switch the chunk to `eval: true` (or just run it interactively) once you're working with the real data. A synthetic stand-in further below lets this whole document render without it.

▼ Code

```
root <- "/Users/nataliiaremezova/Documents/Workspace/BHT/DataViz/GGT-10kEval-90"

aux_files <- list.files(
  root,
  pattern = "\\.auxiliary\\.json$",
  recursive = TRUE,
  full.names = TRUE
)

aux_vars <- c(
  "longitude", "latitude", "elevation", "slope", "aspect",
  "soil_moisture_0_5cm", "soil_moisture_5_15cm", "soil_moisture_15_30cm",
```

```
  paste0("bio", sprintf("%02d", 1:19))
)

read_one <- function(aux_path) {
  key <- basename(aux_path) |> str_remove("\\.auxiliary\\.json$")
  text_path <- file.path(dirname(aux_path), paste0(key, ".text.json"))

  aux   <- fromJSON(aux_path)
  label <- fromJSON(text_path)

  aux_row <- aux[aux_vars]
  aux_row[sapply(aux_row, is.null)] <- NA

  tibble(
    key     = key,
    level0  = label$level0,
    family  = label$level1_family,
    genus   = label$level2_genus,
    species = label$level3_species
  ) |>
    bind_cols(as_tibble(aux_row))
}

df <- map_dfr(aux_files, read_one)

df <- df |>
  rename(
    annual_mean_temp = bio01, mean_diurnal_temp_range = bio02, isothermality = bio03,
    temp_seasonality = bio04, max_temp_warmest_month = bio05, min_temp_coldest_month =
    temp_annual_range = bio07, mean_temp_wettest_quarter = bio08, mean_temp_driest_qua
    mean_temp_warmest_quarter = bio10, mean_temp_coldest_quarter = bio11,
    annual_precipitation = bio12, precip_wettest_month = bio13, precip_driest_month =
    precip_seasonality = bio15, precip_wettest_quarter = bio16, precip_driest_quarter
    precip_warmest_quarter = bio18, precip_coldest_quarter = bio19
  )
```

1.5 Common names for families

Latin family names are precise but not friendly to a non-biologist audience. This builds a lookup table mapping the most common tree families to their everyday English names and joins it onto the data — used in tables/legends from here on instead of raw Latin where it helps readability. Families not in this list (there are 275 in the full dataset) fall back to "Latin name only" rather than `NA`.

▼ Code

```
family_common_names <- tribble(
  ~family,            ~common_name,
  "Pinaceae",         "Pine family",
  "Betulaceae",       "Birch family",
  "Salicaceae",       "Willow family",
  "Fagaceae",         "Beech/Oak family",
  "Rosaceae",         "Rose family",
```

```
  "Fabaceae",       "Legume/Pea family",
  "Proteaceae",     "Protea family",
  "Araliaceae",     "Ginseng/Ivy family",
  "Ulmaceae",       "Elm family",
  "Viburnaceae",    "Viburnum family",
  "Aceraceae",      "Maple family",
  "Sapindaceae",    "Soapberry/Maple family",
  "Oleaceae",       "Olive/Ash family",
  "Juglandaceae",   "Walnut family",
  "Cupressaceae",   "Cypress family",
  "Taxaceae",       "Yew family",
  "Myrtaceae",      "Myrtle/Eucalyptus family",
  "Lauraceae",      "Laurel family",
  "Magnoliaceae",   "Magnolia family",
  "Moraceae",       "Mulberry/Fig family",
  "Anacardiaceae",  "Cashew/Sumac family",
  "Caprifoliaceae", "Honeysuckle family",
  "Ericaceae",      "Heath family",
  "Malvaceae",      "Mallow family",
  "Bignoniaceae",   "Trumpet vine family",
  "Cornaceae",      "Dogwood family",
  "Hamamelidaceae", "Witch-hazel family",
  "Platanaceae",    "Plane/Sycamore family",
  "Cactaceae",      "Cactus family",
  "Asteraceae",     "Daisy/Sunflower family",
  "Myricaceae",     "Bayberry family",
  "Ebenaceae",      "Ebony family",
  "Theaceae",       "Tea family",
  "Meliaceae",      "Mahogany family",
  "Combretaceae",   "Combretum family",
  "Arecaceae",      "Palm family"
)

df <- df |>
  left_join(family_common_names, by = "family") |>
  mutate(
    family_label = if_else(is.na(common_name),
                           family,
                           paste0(common_name, " (", family, ")"))
  )

# Quick check: which of your top families aren't in the lookup yet?
df |> distinct(family, common_name) |> filter(is.na(common_name))
```

```
# A tibble: 0 × 2
# ℹ 2 variables: family <chr>, common_name <chr>
```

📝 The last line is meant to be checked every time you rerun on real data — if it lists families you actually care about, just add a row to `family_common_names` above. The full GlobalGeoTree dataset spans 275 families, so this lookup deliberately only covers commonly known ones rather than trying to be exhaustive.

The most direct answer to "does family X like it cold/hot/dry/wet/high": group by family and summarize. No projection, no dimensionality reduction — just the numbers.

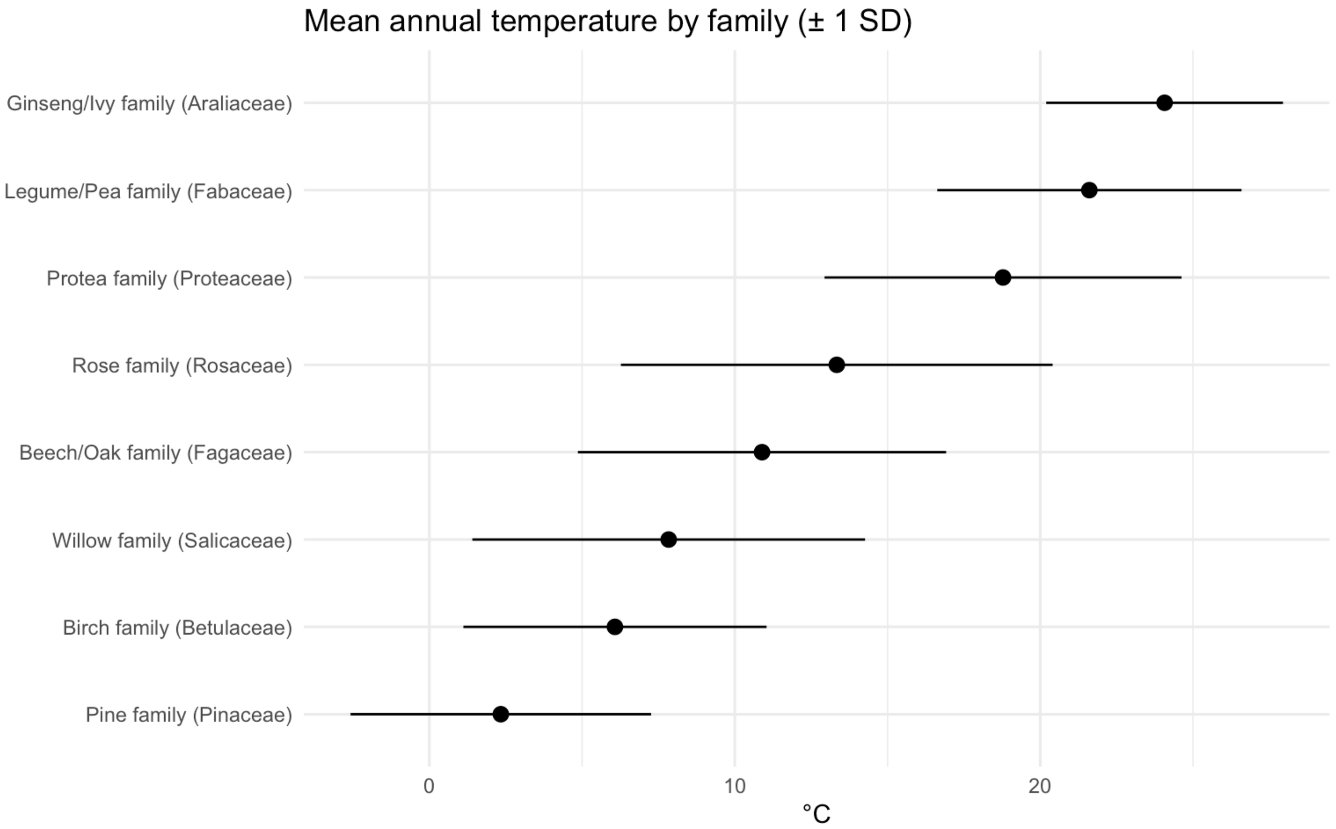
▼ Code

```
top_families <- df |> count(family, sort = TRUE) |> slice_head(n = 10) |> pull(family)

family_niche <- df |>
  filter(family %in% top_families) |>
  group_by(family, family_label) |>
  summarise(
    n = n(),
    mean_temp_C = mean(annual_mean_temp / 10, na.rm = TRUE),
    sd_temp_C = sd(annual_mean_temp / 10, na.rm = TRUE),
    mean_precip_mm = mean(annual_precipitation, na.rm = TRUE),
    sd_precip_mm = sd(annual_precipitation, na.rm = TRUE),
    mean_elevation_m = mean(elevation, na.rm = TRUE),
    sd_elevation_m = sd(elevation, na.rm = TRUE),
    .groups = "drop"
  ) |>
  relocate(family_label, .after = family) |>
  arrange(mean_temp_C)

family_niche
```

```
# A tibble: 8 × 9
  family    family_label     n mean_temp_C sd_temp_C mean_precip_mm sd_precip_mm
  <chr>     <chr>        <int>       <dbl>     <dbl>          <dbl>        <dbl>
1 Pinaceae  Pine family…   491        2.34      4.92           691.         256.
2 Betulace… Birch famil…   516        6.08      4.96           846.         304.
3 Salicace… Willow fami…   495        7.84      6.43           770.         312.
4 Fagaceae  Beech/Oak f…   513       10.9       6.03          1010.         359.
5 Rosaceae  Rose family…   505       13.3       7.07           904.         347.
6 Proteace… Protea fami…   469       18.8       5.84           617.         293.
7 Fabaceae  Legume/Pea …   475       21.6       4.98           443.         229.
8 Araliace… Ginseng/Ivy…   536       24.1       3.88          1785.         493.
# ℹ 2 more variables: mean_elevation_m <dbl>, sd_elevation_m <dbl>
```

▼ Code

```
family_niche |>
  mutate(family_label = fct_reorder(family_label, mean_temp_C)) |>
  ggplot(aes(family_label, mean_temp_C)) +
  geom_pointrange(aes(ymin = mean_temp_C - sd_temp_C, ymax = mean_temp_C + sd_temp_C))
  coord_flip() +
  labs(title = "Mean annual temperature by family (± 1 SD)",
       x = NULL, y = "°C")
```



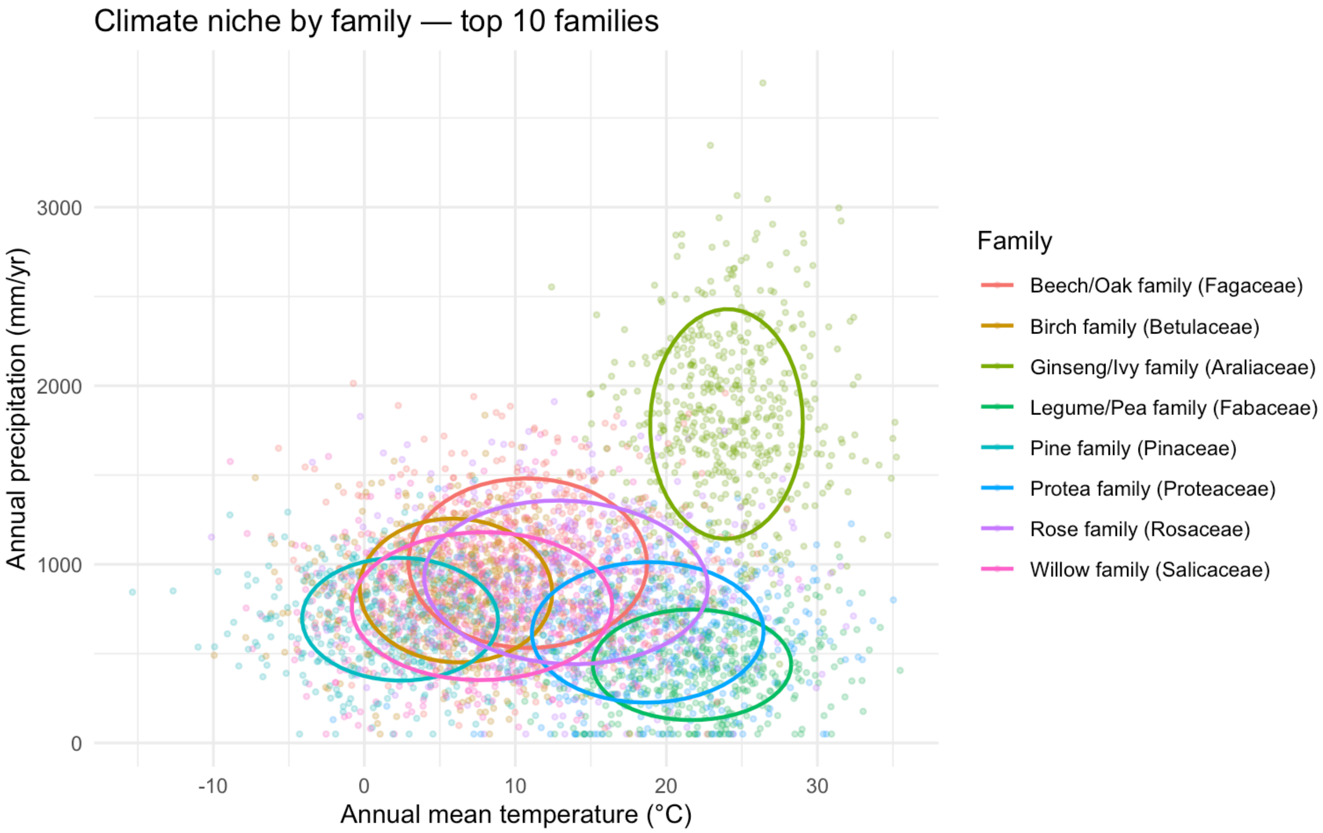
📝 **Reading this plot:** families near the top have the coldest typical climate (point position) and the narrowest/widest range of climates they tolerate (error bar length). A family with a tight error bar around a cold mean is a real candidate for "genuinely cold-adapted family"; a family with a huge error bar spanning both cold and warm temperatures is more likely just widely distributed, not climate-specialized.

3. Climate niche scatter — temperature vs. precipitation

This is the direct, interpretable version of what the PCA biplot was trying to show, except both axes are real units instead of abstract components — closer to a classic Whittaker biome diagram than to a PCA plot.

▼ Code

```
df |>
  filter(family %in% top_families) |>
  ggplot(aes(annual_mean_temp / 10, annual_precipitation, color = family_label)) +
  geom_point(alpha = 0.25, size = 0.8) +
  stat_ellipse(level = 0.68, linewidth = 0.8) +
  labs(title = "Climate niche by family — top 10 families",
       x = "Annual mean temperature (°C)", y = "Annual precipitation (mm/yr)",
       color = "Family")
```



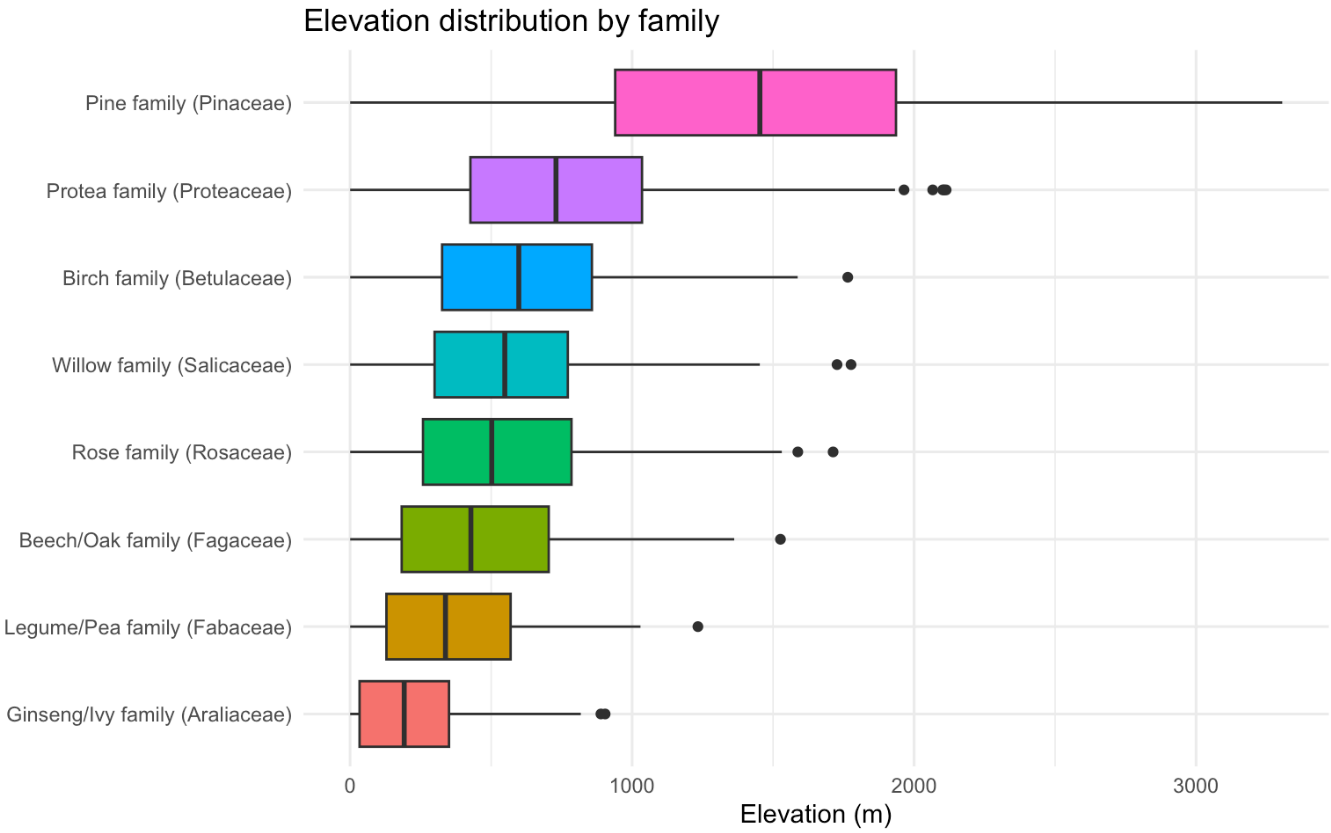
📝 The ellipses (68% confidence region per family) are doing the same job the PCA arrows tried to do, but per-family and in real units. Families whose ellipses barely overlap have a genuinely distinguishable climate niche; families whose ellipses sit on top of each other are climate generalists, or this particular curated sample doesn't capture their niche distinctly. This plot is also where you'd visually spot something like "family X occupies the cold/wet corner" — directly, without inferring it from PC loadings.

4. Elevation by family

Tests the "pines like high/cold places" intuition directly.

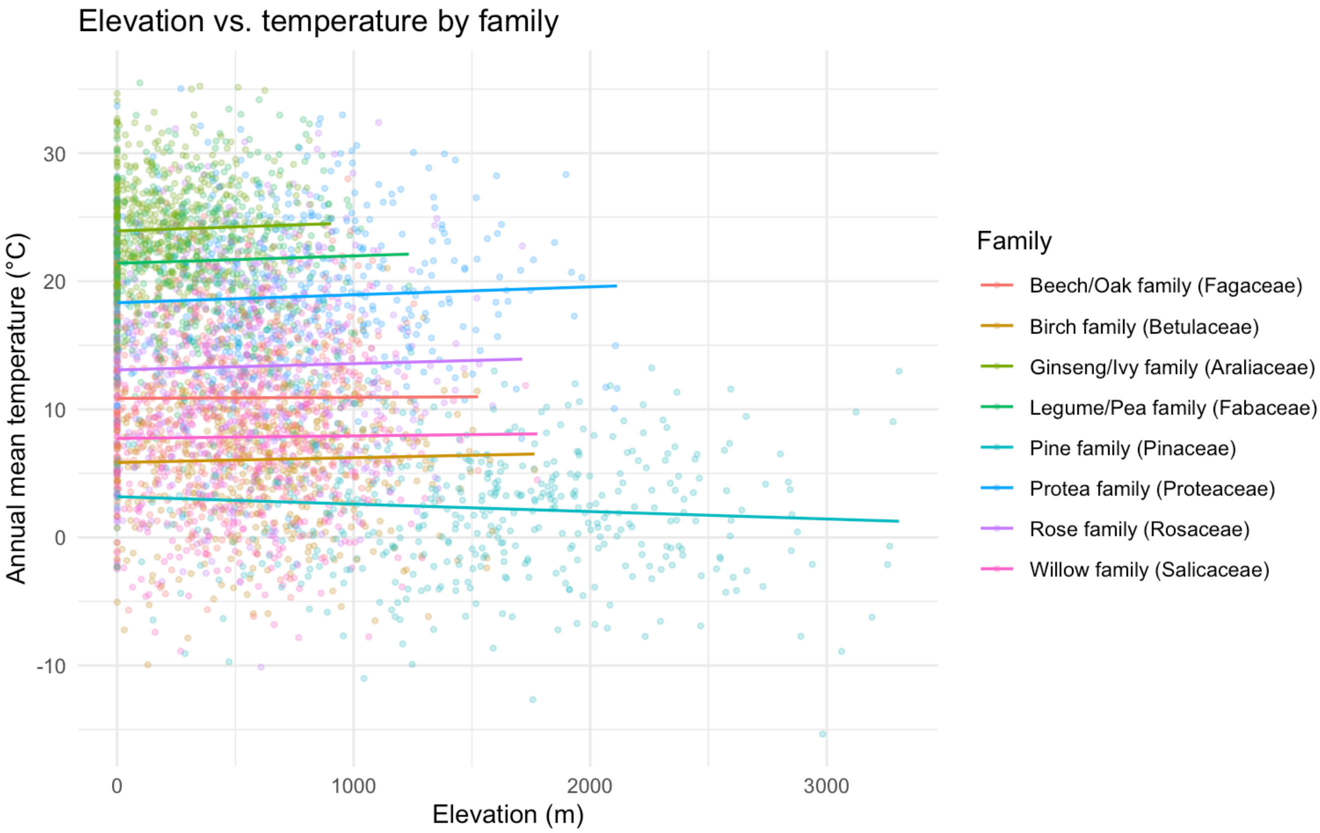
▼ Code

```
df |>
  filter(family %in% top_families) |>
  mutate(family_label = fct_reorder(family_label, elevation, .fun = median)) |>
  ggplot(aes(family_label, elevation, fill = family_label)) +
  geom_boxplot(show.legend = FALSE) +
  coord_flip() +
  labs(title = "Elevation distribution by family", x = NULL, y = "Elevation (m)")
```



▼ Code

```
df |>
  filter(family %in% top_families) |>
  ggplot(aes(elevation, annual_mean_temp / 10, color = family_label)) +
  geom_point(alpha = 0.25, size = 0.8) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.6) +
  labs(title = "Elevation vs. temperature by family",
       x = "Elevation (m)", y = "Annual mean temperature (°C)", color = "Family")
```



📝 The second plot also serves as a sanity check on the data itself: all families should show a *negative* elevation–temperature slope (the physical lapse rate, ~6.5°C per 1000m) regardless of family identity. If some family's line goes the wrong way, that's worth investigating as a possible data or join issue before reading anything ecological into it.

5. Regional map — continental US

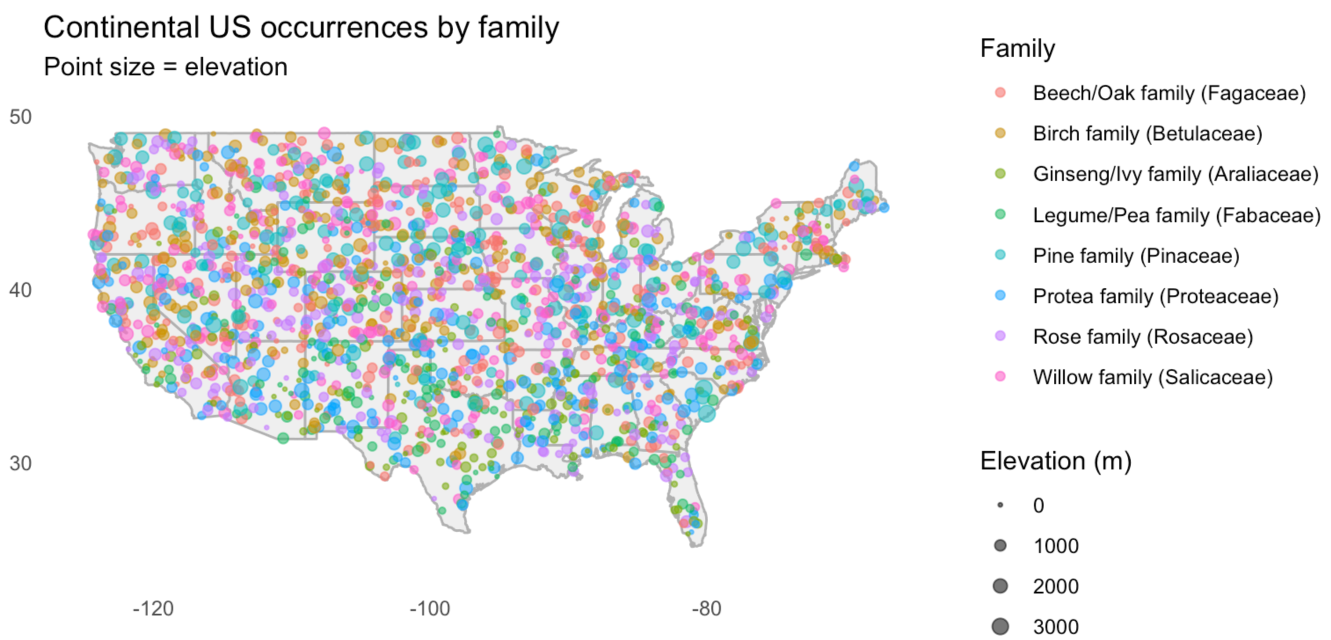
Filtering to a bounding box and plotting real coordinates is far more checkable than an abstract embedding — you can directly compare against known geography (e.g. conifers in the Rockies/Pacific Northwest vs. broadleaf in the East).

🛠 **Fix — clip to actual land, not a rectangle.** A plain lat/lon box (`-125 to -66, 24 to 50`) also catches a strip of southern Canada, northern Mexico, and ocean near the coasts. `maps::map.where()` does a real point-in-polygon test against US state outlines, so points outside the country's actual land area get dropped instead of just clipped to a rectangle. This matters more once you're on real data — with the synthetic placeholder, longitude is pure `runif()` with no relationship to land at all, so even this fix won't make the placeholder maps look geographically meaningful; it'll matter once real coordinates are in.

▼ Code

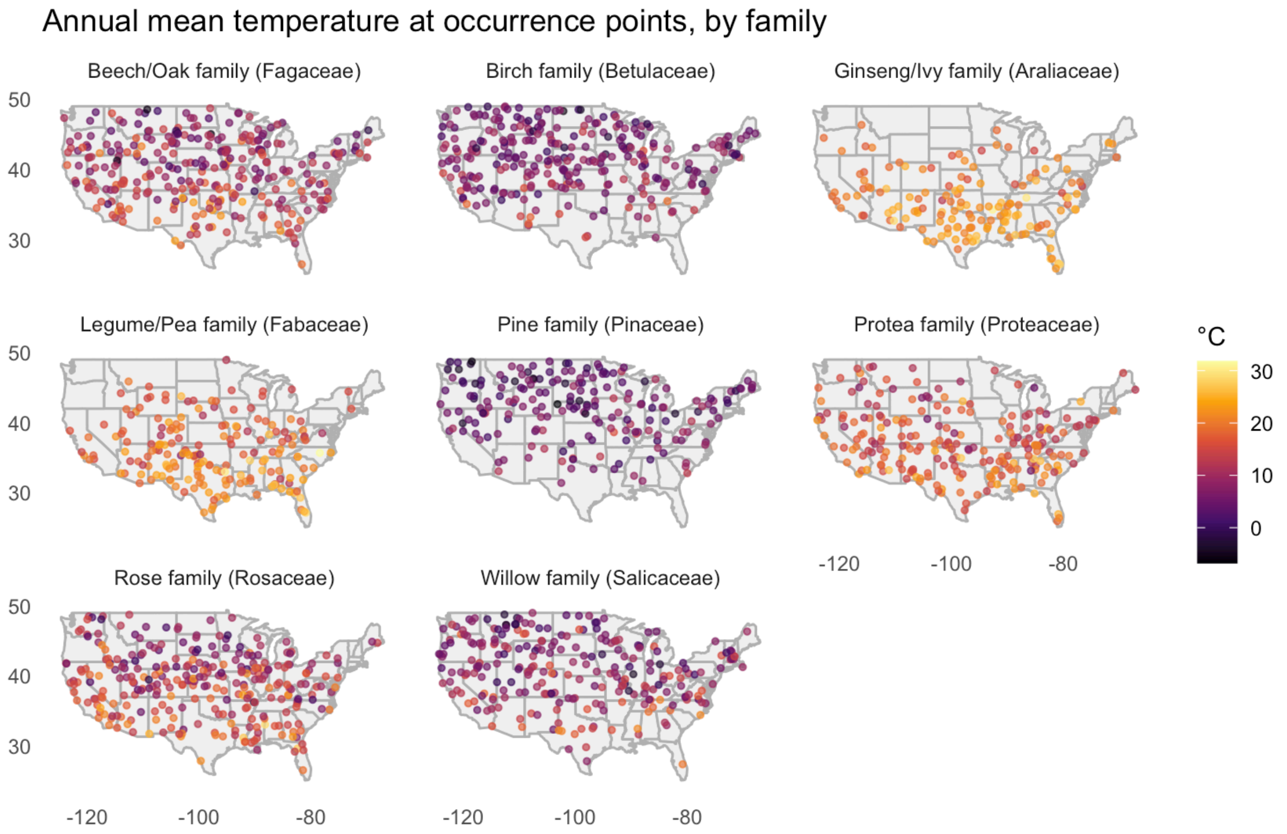
```
# install.packages("maps")  # ships US state borders, no API key needed
library(maps)

us_bbox <- df |>
  filter(longitude > -125, longitude < -66, latitude > 24, latitude < 50) |>
  filter(family %in% top_families) |>
  mutate(us_state = map.where("state", longitude, latitude)) |>
  filter(!is.na(us_state))   # drops ocean / Canada / Mexico points
```

```
ggplot(us_bbox, aes(longitude, latitude)) +
  borders("state", fill = "grey95", colour = "grey70") +
  geom_point(aes(color = family_label, size = elevation), alpha = 0.6) +
  scale_size_continuous(range = c(0.5, 3), name = "Elevation (m)") +
  coord_quickmap(xlim = c(-125, -66), ylim = c(24, 50)) +
  labs(title = "Continental US occurrences by family",
       subtitle = "Point size = elevation",
       x = NULL, y = NULL, color = "Family") +
  theme(panel.grid = element_blank())
```



▼ Code

```
ggplot(us_bbox, aes(longitude, latitude)) +
  borders("state", fill = "grey95", colour = "grey70") +
  geom_point(aes(color = annual_mean_temp / 10), alpha = 0.7, size = 1) +
  scale_color_viridis_c(option = "inferno", name = "°C") +
  coord_quickmap(xlim = c(-125, -66), ylim = c(24, 50)) +
  facet_wrap(~family_label) +
  labs(title = "Annual mean temperature at occurrence points, by family",
       x = NULL, y = NULL) +
  theme(panel.grid = element_blank())
```



📝 **Reading these maps.** The first map is the most intuitive check — does a cold-climate family actually cluster in the Rockies/Pacific Northwest/New England rather than the Gulf Coast? The faceted second map controls for the fact that "where points are" and "how warm those points are" can both vary — a family could be regionally clustered for reasons unrelated to temperature (sampling effort, data source), so check whether the color (temperature) pattern within each facet looks geographically sensible (warmer toward the south/coasts, cooler at elevation/north) before attributing a pattern to the family's climate preference specifically.

⚠️ **If you're still seeing points scattered uniformly with no geographic structure (e.g. in the ocean, or no visible north/south temperature gradient) after rerunning this**, that's a sign the `load-data` chunk is still set to `eval: false` and you're looking at the synthetic placeholder rather than real GlobalGeoTree coordinates — flip that chunk on and delete the `simulate-data` chunk below it.

⚠️ Note on cacti: GlobalGeoTree is **tree occurrences only**, so true cacti (Cactaceae) are unlikely to appear — most cacti aren't trees. If you're looking for the "loves it hot and dry" analogue in this dataset, check which genera/families actually show up in the arid Southwest in the map above (candidates: *Prosopis* (mesquite), *Acacia*-type Fabaceae genera, or *Yucca* if classified as tree-form here) rather than assuming cacti specifically will be present.

6. A direct, supervised check: does climate predict family?

Rather than reading tea leaves from a 2D projection, fit a small model and look at which variables it actually uses. This converts "I think I see clustering" into a number.

▼ Code

```
# install.packages("randomForest")
library(randomForest)
```

```
rf_data <- df |>
  filter(family %in% top_families) |>
  select(family, elevation, slope, annual_mean_temp, temp_seasonality,
         annual_precipitation) |>
  mutate(family = factor(family)) |>
  na.omit()

set.seed(1)
rf <- randomForest(family ~ ., data = rf_data, ntree = 300, importance = TRUE)

# class.error keyed by Latin name (model's factor levels) — join back to
# common names for a readable readout
tibble(family = rownames(rf$confusion), class_error = rf$confusion[, "class.error"]) |
  left_join(family_common_names, by = "family") |>
  arrange(class_error)

varImpPlot(rf, main = "Variable importance for predicting family from climate")
```

📝 **How to read this.** `class.error` per family tells you, concretely, which families *are* climate-distinguishable in this data (low error) and which aren't (high error, near the ~90% chance level for 10 balanced classes) — a much more direct readout than squinting at cluster overlap in a scatter plot. The importance plot tells you which of elevation, temperature, seasonality, or precipitation is actually doing the distinguishing work, which directly answers "is it the cold, the dryness, or the altitude that matters for this family" rather than leaving you to guess from a PCA loading.

7. Summary / where to go next

- Skip PCA/t-SNE as the entry point for this question — they compress variance, not climate-family relationships specifically, and on this dataset the variance they compress doesn't line up cleanly with family.
- The niche table (Section 2) and scatter (Section 3) directly answer "does family X prefer cold/hot/wet/dry," in real units, per family.
- The elevation analysis (Section 4) is the cleanest test of a "pines like high/cold places"-style hypothesis, and includes a built-in sanity check (the lapse rate) for catching data issues.
- The US map (Section 5) is the most interpretable visual — it can be directly compared against known biogeography (conifers in the mountains/north, broadleaf in the east, etc.), which an abstract embedding can't offer.
- The random forest check (Section 6) turns "I think I see structure" into a per-family error rate and a ranked list of which climate variables matter — worth running before writing up any claims about which families are climate-specialized.
- Natural extension: repeat Sections 2–5 on the 300- or 900-species eval splits for more samples per family, and restrict the US map to a single well-known contrast pair (e.g. Pinaceae vs. a warm-climate broadleaf family) for the clearest possible illustration.